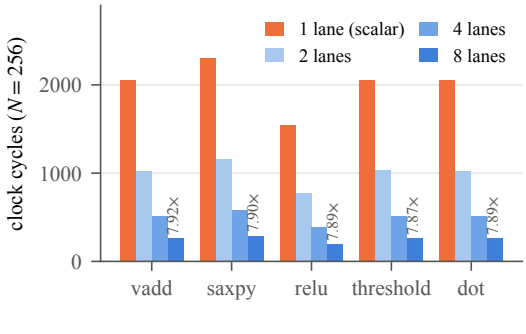
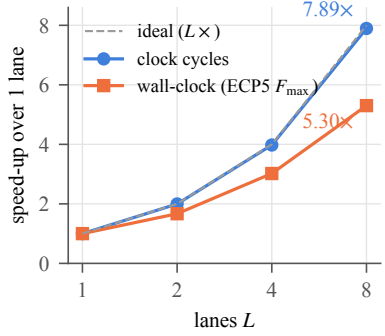
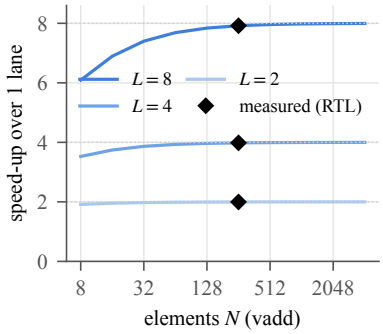

clock cycles ($N = 256$)
2000
1000
0
1 lane (scalar)
2 lanes
4 lanes
8 lanes
7.92×
7.90×
7.89×
7.87×
7.89×
vadd
saxpy
relu
threshold
dot


speed-up over 1 lane
8
6
4
2
0
ideal ($L\times$)
clock cycles
wall-clock (ECP5 F_{max})
7.89×
5.30×
1
2
4
8
lanes L


speed-up over 1 lane
8
6
4
2
0
$L = 8$
$L = 4$
$L = 2$
measured (RTL)
8
32
128
512
2048
elements N (vadd)